

Package ‘spaCraft’

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Title spaCraft: Analysis-aware power calculation and sample size planning for multi-sample spatial transcriptomics

Version 1.0.1

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Description Provides a pilot-based framework for statistical power analysis and sample-size determination in comparative, multi-sample spatial transcriptomics studies. A cohort-level generative model is learned from pilot data through three layers: a baseline-anchored Binomial-logit model for domain composition, a Fisher-Gaussian kernel mixture model for spatial geometry (Mukhopadhyay, Li and Dunson (2020) <[doi:10.1111/rssb.12390](https://doi.org/10.1111/rssb.12390)>), and a Gaussian-process expression model fitted with nearest-neighbor approximations (Saha and Datta (2018) <[doi:10.1002/sta4.184](https://doi.org/10.1002/sta4.184)>). Synthetic cohorts are generated under user-specified effect sizes, spatial domains are recovered with a pilot-guided extension of 'BANKSY' (Singhal et al. (2024) <[doi:10.1038/s41588-024-01664-3](https://doi.org/10.1038/s41588-024-01664-3)>), and endpoint-specific power is estimated under a full generate-recover-test pipeline for a spatially adjusted differential-expression test and a baseline-anchored compositional test, both using a minimum-effect 'TREAT' margin (McCarthy and Smyth (2009) <[doi:10.1093/bioinformatics/btp053](https://doi.org/10.1093/bioinformatics/btp053)>). Monte-Carlo power estimates are summarized with monotone shape-constrained additive models to recommend per-group sample sizes. The framework supports both replicated and single-pilot designs and has been evaluated across the 'Visium', 'Visium HD', and 'Stereo-seq' platforms.

Depends R (>= 4.1.0)

License GPL (>= 3)

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 8.0.0

LinkingTo Rcpp,
RcppArmadillo

Imports BRISC, clue, Directional, dplyr, FNN, ggplot2, MASS, Matrix,
matrixStats, mclust, MCMCpack, methods, patchwork, Rcpp, scam, stats, limma

Suggests data.table, fields, ggrepel, knitr, mgcv, RANN, readr, rmarkdown,
RSpectra, Seurat, SingleCellExperiment, SpatialExperiment, spatialLIBD,
stringr, testthat (>= 3.0.0)

VignetteBuilder knitr

Config/testthat/edition 3

biocViews Software, SingleCell, Spatial, Transcriptomics, GeneExpression,
DifferentialExpression, Clustering, Bayesian

Collate 'LOR.R'

'PowerEvaluation.R'

'RcppExports.R'

'SaLFC.R'

'createspaCraftObject.R'

'estimation.R'

'featureSelection.R'

'fn_FGEM_aux.R'

'fn_aux_base.R'

'lambda_tuning.R'

'pBANSKY.R'

'rearrangement.R'

'simulatespaCraft.R'

'spaCraft-package.R'

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as_powerSaLFC	<i>Tag a SaLFC power grid as class "powerSaLFC"</i>
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Description

Tag a SaLFC power grid as class "powerSaLFC"

Usage

```
as_powerSaLFC(x, alpha = 0.05)
```

Arguments

x	Data frame from evaluatePowerSaLFC.
alpha	Numeric. The (FDR) level used, stored for display.

Value

x with class c("powerSaLFC", ...) and an alpha attribute.

createspaCraftObject	<i>Create spaCraft Object</i>
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Description

Creates a spaCraft object from a pre-processed list of pilot data. If the dataset contains only single replicates (K=1) globally or per group, this function can automatically generate pseudo-replicates via minor random perturbations to ensure downstream multi-sample estimators function correctly.

Usage

```
createspaCraftObject(
  pilot_list,
  generate_pseudo_reps = TRUE,
  noise_level = 0.05
)
```

Arguments

pilot_list	A list of processed samples. Each element must contain counts (genes x spots), coords (with columns 'x', 'y', 'domain'), and group.
generate_pseudo_reps	Logical. If TRUE, expands any group with K=1 to K=3 by adding minor spatial and count perturbations. Default is TRUE.
noise_level	Numeric. Fractional uniform noise added to counts for pseudo-replicates. Default is 0.05 (+/- 5%).

Value

A spaCraft object ready for parameter estimation.

estimateCompositionParams

Estimate Composition Parameters (Multi-Sample)

Description

Estimates parameters for the composition model via Binomial GLM and empirical pilot moments. Requires $K > 1$ replicates per group.

Usage

```
estimateCompositionParams(
  object,
  target_domain,
  reference_domain,
  prob_floor = 1e-06,
  verbose = TRUE
)
```

Arguments

object	A spaCraft object containing pilot_data.
target_domain	Character. Target domain name.
reference_domain	Character. Reference domain name.
prob_floor	Numeric. Probability clamp for numerical safety.
verbose	Logical. Print progress.

Value

Updated spaCraft object with params_composition.

estimateExpressionParams

Estimate Expression Parameters (Multi-Sample Optimized)

Description

Estimates gene expression parameters across multiple samples. Uses Rcpp/OpenMP to pre-compute biological variance, grand means, and residuals, massively speeding up the spatial estimation loop.

Usage

```
estimateExpressionParams(
  object,
  target_domain,
  reference_domain,
  genes_to_use = NULL,
  n_neighbors = 15,
  n_cores = 1,
  verbose = TRUE
)
```

Arguments

object	A spaCraft object.
target_domain	Character. Target domain name.
reference_domain	Character. Reference domain name.
genes_to_use	Character vector or NULL. Genes to evaluate.
n_neighbors	Integer. Number of neighbors for spatial modeling (default: 15).
n_cores	Integer. Number of cores for parallel processing.
verbose	Logical. Print progress messages.

Value

Updated spaCraft object with params_expression populated.

Examples

```
## Not run:
data("mini_obj_features", package = "spaCraft")
spaCraft_obj <- estimateExpressionParams(
  object = mini_obj_features,
  target_domain = "WM",
  reference_domain = "Layer6",
  genes_to_use = sets$G_svg,
  n_cores = 8
)

## End(Not run)
```

estimateGeometryParams

Estimate Spatial Geometry Parameters (Multi-Sample)

Description

Estimates parameters for domain geometry via FGKMM pooling across multiple samples.

Usage

```
estimateGeometryParams(
  object,
  M_candidates = 3:6,
  iter_max = 500,
  tol = 0.01,
  n_cores = 1,
  verbose = TRUE,
  min_spots = 30,
  trim_outliers = TRUE,
  trim_k = 10L,
  trim_quantile = 0.98,
  gc_each_fit = FALSE,
  store_criteria = TRUE,
  worker_sources = NULL,
  max_pool_spots = 5000L
)
```

Arguments

object	A spaCraft object.
M_candidates	Integer vector. Candidate M values for FGKMM.
iter_max	Integer. Max EM iterations.
tol	Numeric. EM convergence tolerance.
n_cores	Integer. Number of PSOCK cores.
verbose	Logical. Print progress.
min_spots	Integer. Minimum spots per domain per sample.
trim_outliers	Logical. Density-based outlier trimming before FGKMM fitting.
trim_k	Integer. KNN for density estimation.
trim_quantile	Numeric in (0,1).
gc_each_fit	Logical. Trigger garbage collection each fit.
store_criteria	Logical. Store BIC criteria table.
worker_sources	Character vector. Optional R scripts to source on PSOCK workers.
max_pool_spots	Integer. Max pooled centered spots for M selection.

Value

Updated spaCraft object with params_geometry.

estimate_lfc_threshold_from_pilot

Estimate Gene-Specific SaLFC TREAT Thresholds from Pilot Data

Description

Fast plug-in estimator (no bootstrap) of gene-specific LFC thresholds τ_g for the TREAT-style SaLFC test, using only pilot data. SaLFC is run once on the pilot samples with `lfc_threshold = 0`; τ_g is then defined from the pilot effect size and its uncertainty in a direction-aware way:

$$\tau_g = \max(\text{effect_ref}_g, z_q \cdot \text{se}_g),$$

where `effect_ref` depends on alternative and $z_q = \Phi^{-1}(q)$.

This module is normally invoked automatically by `SaLFC(..., lfc_threshold = "pilot")`, but it can also be called directly and its result passed as `lfc_threshold`.

Usage

```
estimate_lfc_threshold_from_pilot(
  object,
  genes,
  target_domain = "WM",
  reference_domain = "Layer6",
  rho_TR = 0,
  adjust_method = "bonferroni",
  test_method = "t",
  alternative = c("greater", "less", "two.sided"),
  q = 0.75,
  min_tau = 0,
  cap_effect = NULL,
  max_spots_per_domain = 5000L,
  verbose = TRUE,
  return_table = FALSE,
  test_fn = NULL,
  B = 200,
  shrink_to_global = 0.25,
  seed = 1
)
```

Arguments

<code>object</code>	A spaCraft object with <code>@pilot_data</code> .
<code>genes</code>	Character vector of (null) genes, e.g. the stable/null set.
<code>target_domain</code>	Character. Target domain label.
<code>reference_domain</code>	Character. Reference domain label.
<code>rho_TR</code>	Numeric. Assumed Target-Reference biological correlation (< 1).
<code>adjust_method</code>	Multiple-testing method passed through to SaLFC.
<code>test_method</code>	SaLFC group comparison: "t" or "wilcoxon".

alternative	Test direction: "greater" (default), "less", "two.sided".
q	Numeric in (0,1). Standard-normal quantile level for z_q .
min_tau	Numeric. Lower bound for τ_g .
cap_effect	Numeric scalar, named numeric vector, or NULL. Optional upper cap on τ_g (per gene if named).
max_spots_per_domain	Integer. Per-sample spot cap before running SaLFC on the pilot (bounds memory). The global RNG state is restored on exit.
verbose	Logical.
return_table	Logical. If TRUE return a data.frame, else a named vector.
test_fn	Function. The SaLFC test to run on the pilot. If NULL, resolves SaLFC by name. When invoked via <code>SaLFC(..., lfc_threshold = "pilot")</code> , the running test function is passed automatically, so renaming it does not break this module.
B, shrink_to_global, seed	Deprecated and ignored; retained for backward compatibility.

Value

A named numeric vector τ_g carrying an `attr(, "global_tau")`, or a data.frame when `return_table = TRUE`.

See Also

[SaLFC](#)

estimate_lor_tau_from_pilot

Estimate the Composition TREAT Threshold (tau) from Pilot Data

Description

Plug-in estimator of the scalar TREAT threshold τ for the sample-level composition test [LOR](#), on the log-odds-ratio (LOR) scale. The threshold is the larger of a scientific anchor and a finite-sample noise floor:

$$\tau = \max(\tau_{\text{sci}}, \tau_{\text{noise}}, \text{min_tau}).$$

Scientific anchor τ_{sci} (from fitted values; no pilot simulation). The direction-aware log-odds distance between a minimum meaningful CASE target proportion `target_prop_min` (written p^*) and the fitted baseline proportion p_0 :

$$\tau_{\text{sci}} = |\text{logit}(p^*) - \text{logit}(p_0)|.$$

Under `null_type = "treat"` the baseline is $p_0 = \text{plogis}(\beta_0 + \beta_1)$ (the fitted case mean, stored as `mu1_hat`), because the test centers at $\delta_0 = \beta_1$; under `"two_sample"` it is $p_0 = \text{plogis}(\beta_0)$ (stored as `mu0_hat`). This term is 0 when `target_prop_min = NULL`.

Noise floor τ_{noise} (optional; finite-sample). A pilot band $t_{q,df}$ *se*: LOR is run once on the pilot (centered null, `tau = 0`) purely to read the Welch standard error `se` and degrees of freedom `df`; the multiplier is `qt(q, df)`, or `qnorm(q)` when `df` is unavailable (e.g. `wilcoxon`). Set `use_noise_floor = FALSE` to drop this term and obtain τ as a deterministic function of the fitted values alone (no pilot run). Returns one scalar per (target, reference) pair. Normally invoked automatically by `LOR(..., tau = "pilot")` or `evaluatePowerLOR(..., tau = "auto")`; can also be called directly. The pilot run uses the ground-truth `coords$domain` (pilot data has no `hat_d`).

Usage

```
estimate_lor_tau_from_pilot(
  object,
  target_domain = "WM",
  reference_domain = "Layer6",
  null_type = c("treat", "two_sample"),
  test_method = c("t", "wilcoxon"),
  alternative = c("greater", "less", "two.sided"),
  target_prop_min = NULL,
  q = 0.5,
  use_noise_floor = TRUE,
  min_tau = 0,
  cap_effect = NULL,
  eps = 0.5,
  verbose = TRUE,
  test_fn = NULL
)
```

Arguments

object	A spaCraft object with @params_composition (and, when use_noise_floor = TRUE, @pilot_data).
target_domain, reference_domain	Character domain labels.
null_type	"treat" (default) or "two_sample". Selects the fitted baseline for the anchor: μ_{1_hat} (treat) or μ_{0_hat} (two_sample). NB: LOR/evaluatePowerLOR force $\tau = 0$ under "two_sample", so in that pipeline τ matters only for "treat".
test_method	"t" (default) or "wilcoxon". Only "t" yields an se for the noise floor; "wilcoxon" uses the anchor alone (noise term = 0).
alternative	"greater" (default), "less", or "two.sided". Sets the sign convention of the anchor: greater keeps only positive departures of target_prop_min above the baseline, less only negative, two.sided the absolute gap.
target_prop_min	Numeric in (0,1) or NULL (default). Minimum meaningful CASE target proportion driving τ_{sci} ; NULL disables the anchor ($\tau_{sci} = 0$).
q	Numeric in (0,1). Quantile level for the noise-floor multiplier.
use_noise_floor	Logical (default TRUE). Add the pilot noise floor τ_{noise} . FALSE returns $\max(\tau_{sci}, \min_tau)$ with no pilot run.
min_tau	Numeric. Lower bound for τ .
cap_effect	Numeric or NULL. Optional upper cap on τ .
eps	Numeric > 0. Continuity correction forwarded to LOR.
verbose	Logical.
test_fn	Function. The composition test to run on the pilot (used only when use_noise_floor = TRUE). If NULL, resolves LOR by name. When invoked via LOR(..., tau = "pilot"), the running test function is passed automatically.

Value

A numeric scalar τ ($\geq \min_tau$).

See Also

[LOR](#), [evaluatePowerLOR](#)

evaluatePowerLOR

Evaluate Composition (LOR) Power over a Grid

Description

Grid simulation over sample sizes (**K**) and composition effect sizes to evaluate the power and empirical size of the sample-level log-odds-ratio test [LOR](#). Per replicate the pipeline is `simulatespaCraft()` -> `pBANSKY()` -> `LOR()`. The effect size is the case-vs-control shift in the `target_domain` abundance, injected through the simulator's composition scenario (`delta_pp`, in percentage points, or `target_prop_case`); expression DE is switched off (`DE_lfc = 0`, `de_genes = character(0)`) so the swept signal is purely compositional.

Unlike the per-gene SaLFC test, LOR returns a SINGLE test per (target, reference) pair, so there is no multiple testing within a replicate: **power = rejection rate** across replicates at each (**K**, effect), and the effect = 0 row is the **empirical size**. The composition TREAT threshold τ does not depend on the effect size, so under `null_type = "treat"` the pilot τ is estimated ONCE (a scalar) and reused. A progress bar is shown and simulation errors are skipped.

Usage

```
evaluatePowerLOR(
  object,
  K_grid,
  delta_grid,
  n_sim = 30,
  seed = 1,
  genes,
  target_domain,
  reference_domain,
  effect_type = c("target_prop_case", "delta_pp"),
  eval_on = c("hat_d", "domain"),
  null_type = c("treat", "two_sample"),
  tau = "auto",
  tau_q = 0.5,
  target_prop_min = NULL,
  tau_args = list(),
  test_method = c("t", "wilcoxon"),
  alternative = c("greater", "two.sided", "less"),
  alpha = 0.05,
  eps = 0.5,
  rearrange = FALSE,
  jitter_frac = 0.3,
  k_nn = 50L,
  n_cores = 1L,
  scenario_base = list(kappa_m = 1, kappa_bio = 1, delta_rho = 1, lambda_cond = 0.5),
  clustering_args = list(lambda = 0.5, k_neighbors = 20, scale_pilot_coords = FALSE,
    scale_sim_coords = FALSE, hard_guidance = FALSE, n_pcs = 20, do_hungarian = TRUE,
    use_G = TRUE, balance_mode = "energy_proportional"),
```

```
tuning_obj = NULL,
...
)
```

Arguments

object	A spaCraft object (fitted: pilot, params, @params_composition).
K_grid	Numeric vector of samples per group (e.g. seq(3, 8)).
delta_grid	Numeric vector of composition effect sizes. When effect_type = "delta_pp" these are percentage-point shifts of the target proportion (0 = null); when "target_prop_case" they are the absolute case-group target proportions in (0, 1).
n_sim	Integer. Replications per grid point.
seed	Integer. Master seed.
genes	Character vector of genes to simulate (e.g. sets\$G_svg); needed only so pBANSKY has counts to cluster. Expression DE is off.
target_domain, reference_domain	Domain labels.
effect_type	"delta_pp" or "target_prop_case" (default).
eval_on	Label column LOR counts by: "hat_d" (recovered clusters, default) or "domain" (ground-truth oracle).
null_type	"treat" (default; centered at the pilot baseline $\delta_0 = \beta_1$ with margin τ) or "two_sample" ($H_0 : \bar{L}_1 = \bar{L}_0, \tau$ forced to 0).
tau	"auto" (default; pilot plug-in scalar via estimate_lor_tau_from_pilot, built from null_type, tau_q, target_prop_min and tau_args) or a numeric scalar. Used only when null_type = "treat".
tau_q	Numeric in (0,1). Quantile level for the auto τ noise-floor multiplier $t_{q,df}$. Default 0.5. NB: tau_q = 0.5 gives $t_{0.5,df} = 0$, hence a zero noise floor; with target_prop_min = NULL the auto τ then collapses to 0.
target_prop_min	Numeric in (0,1) or NULL (default). Forwarded to estimate_lor_tau_from_pilot() to set the scientific-anchor τ_{sci} = the (direction-aware) log-odds gap of target_prop_min from the fitted case baseline $\mu_{1_hat} = \text{plogis}(\beta_0 + \beta_1)$ (under null_type = "treat"). NULL leaves the auto τ as the pilot noise floor alone. Only used when tau = "auto".
tau_args	Named list of extra arguments forwarded to estimate_lor_tau_from_pilot() for the auto τ , e.g. list(use_noise_floor = FALSE) for a pure fitted anchor, or list(cap_effect = 0.5, min_tau = 0.02, q = 0.9). Entries here override the defaults built from null_type/tau_q/target_prop_min.
test_method	"t" (Welch, default) or "wilcoxon".
alternative	"greater" (default), "two.sided", or "less".
alpha	Numeric. Rejection threshold.
eps	Numeric > 0. Continuity correction forwarded to LOR.
rearrange	Logical. Morph synthetic coords onto pilot geometry first. Default FALSE: LOR counts by label and is invariant to the spatial rearrangement, so this only costs time.
jitter_frac, k_nn	Rearrangement tuning (used only when rearrange).

n_cores	Integer. Cores for simulate/cluster.
scenario_base	List. Base scenario; any delta_pp, target_prop_case, or DE_lfc here are dropped (they are controlled by the sweep).
clustering_args	List passed to pBANSKY (must contain lambda). NB: for composition power the recovered hat_d must preserve per-sample target abundance, so avoid size-equalizing balance modes.
tuning_obj	Optional spaCraftTuning object from tuneSpaDesignLambdas . When supplied, lambda_cond (scenario) and lambda (clustering_args) are taken <i>per effect size</i> from optimal_params\$LOR\$by_effect (matched on target_prop_case; nearest value with a warning if no exact match). Only applied when effect_type = "target_prop_case". Default NULL.
...	Forwarded simulate tuning: graph_k, gp_rank, gp_jitter, cond_k_nn.

Value

A data.frame of per-replicate rejection rows, tagged with class "powerLOR" (see [print.powerLOR](#), [summary.powerLOR](#), [plotPowerCurveLOR](#)).

See Also

[evaluatePowerSaLFC](#), [LOR](#), [estimate_lor_tau_from_pilot](#), [tuneSpaDesignLambdas](#)

evaluatePowerSaLFC	<i>Evaluate SaLFC Power over a Grid</i>
--------------------	---

Description

Grid simulation over sample sizes (K) and effect sizes (DE_lfc) to evaluate the power and FDP of the SaLFC endpoint. Per replicate the pipeline is `simulatespaCraft() -> pBANSKY() -> rearrangeSyntheticToPilot() -> SaLFC()`. Synthetic coordinates are morphed onto the pilot domain geometry by `eval_domain_col` ("hat_d" by default), and SaLFC selects target/reference by the SAME column, so the spatial variance τ^2 is computed on a consistent basis. Pilot TREAT thresholds (`lfc_threshold = "auto"`) are estimated once per DE_lfc and shared across replicates. A progress bar is shown and simulation errors are skipped gracefully.

Usage

```
evaluatePowerSaLFC(
  object,
  K_grid,
  lfc_grid,
  n_sim = 30,
  seed = 1,
  G_svg,
  G_null,
  G_spike,
  target_domain,
  reference_domain,
  eval_domain_col = c("hat_d", "domain"),
```

```

    rearrange = TRUE,
    jitter_frac = 0.3,
    k_nn = 100L,
    test_method = c("t", "wilcoxon"),
    adjust_method = c("bonferroni", "BH"),
    alternative = c("greater", "less", "two.sided"),
    alpha_fdr = 0.05,
    lfc_threshold = "auto",
    lfc_q = 0.5,
    spatial_mode = c("original", "basis"),
    n_cores = 1L,
    scenario_base = list(target_prop_case = NULL, delta_pp = NULL, kappa_m = 1, kappa_bio =
      1, delta_rho = 1, lambda_cond = 0.5),
    clustering_args = list(lambda = 0.5, k_neighbors = 20, scale_pilot_coords = FALSE,
      scale_sim_coords = FALSE, hard_guidance = FALSE, n_pcs = 20, do_hungarian = TRUE,
      use_G = TRUE, balance_mode = "energy_proportional"),
    tuning_obj = NULL,
    ...
  )

```

Arguments

<code>object</code>	A spaCraft object (fitted: pilot + params).
<code>K_grid</code>	Numeric vector of samples per group (e.g. <code>seq(3, 8)</code>).
<code>lfc_grid</code>	Numeric vector of effect sizes (DE_lfc, on the count-model log scale used by <code>simulatespaCraft</code>).
<code>n_sim</code>	Integer. Replications per grid point.
<code>seed</code>	Integer. Master seed.
<code>G_svg</code>	Character. All genes to simulate (<code>sets\$G_svg</code>).
<code>G_null</code>	Character. Null genes to evaluate (<code>sets\$G_null</code>).
<code>G_spike</code>	Character. True spike-in DE genes (<code>sets\$G_spike</code>).
<code>target_domain, reference_domain</code>	Domain labels.
<code>eval_domain_col</code>	Label column for morphing AND testing: "hat_d" (recovered clusters, default) or "domain" (ground truth).
<code>rearrange</code>	Logical. Morph synthetic blobs onto pilot geometry first.
<code>jitter_frac, k_nn</code>	Rearrangement tuning (used when <code>rearrange</code>).
<code>test_method</code>	"t" or "wilcoxon".
<code>adjust_method</code>	"BH" or "bonferroni" (default).
<code>alternative</code>	SaLFC direction: "greater" (default), "less", "two.sided".
<code>alpha_fdr</code>	Numeric. Rejection threshold for power/FDP.
<code>lfc_threshold</code>	"auto" (pilot plug-in), a scalar, or a named vector.
<code>lfc_q</code>	Numeric. Quantile for the auto threshold.
<code>spatial_mode</code>	"original" or "basis".
<code>n_cores</code>	Integer. Cores for simulate/cluster (keep modest under nesting).

scenario_base	List. Base scenario (composition kept neutral by default so the swept effect is purely expression DE).
clustering_args	List. Passed to pBANSKY (must contain lambda).
tuning_obj	Optional spaCraftTuning object from tuneSpaDesignLambdas . When supplied, the conditional-texture weight lambda_cond (in the simulation scenario) and the clustering weight lambda (in clustering_args) are taken <i>per effect size</i> from the matching row of optimal_params\$SaLFC\$by_effect (matched on DE_lfc; nearest value with a warning if there is no exact match), overriding the values in scenario_base/clustering_args. Default NULL.
...	Forwarded simulate tuning: graph_k, gp_rank, gp_jitter, cond_k_nn.

Value

A data.frame of per-replicate power/FDP rows, tagged with class "powerSaLFC" (see [print.powerSaLFC](#), [summary.powerSaLFC](#), [plotPowerCurveSaLFC](#)).

See Also

[evaluatePowerLOR](#), [SaLFC](#), [tuneSpaDesignLambdas](#)

featureSelection

Select Domain-Informative Genes via Multi-Context Meta-Analysis

Description

Identifies robust marker genes by evaluating spatial expression across multiple biological contexts (e.g., NAT vs. CRC). This meta-analysis mitigates representation bias, ensuring the final feature space reflects the full transcriptomic heterogeneity.

Usage

```
featureSelection(
  object,
  rep_sample_ids = NULL,
  logfc_cutoff = 0.5,
  mean_in_cutoff = 1,
  max_num_gene = 20,
  n_cores = 1,
  combine_method = c("union", "intersection"),
  verbose = TRUE
)
```

Arguments

object	A spaCraft object.
rep_sample_ids	Character vector of representative samples. At least one sample per condition should be provided. Defaults to the first sample if NULL.
logfc_cutoff	Numeric. Minimum absolute log fold change (default: 0.5).
mean_in_cutoff	Numeric. Minimum mean log-expression (default: 1.0).

max_num_gene Integer. Max top genes to extract per domain (default: 20).
 n_cores Integer. CPU cores for parallel processing (default: 1).
 combine_method Character. "union" (default) or "intersection".
 verbose Logical. Print progress.

Value

Updated spaCraft object with integrated features.

Examples

```
## Not run:
data("mini_spaCraft_obj", package = "spaCraft")
spaCraft_obj <- featureSelection(
  object = mini_spaCraft_obj,
  logfc_cutoff = 0.5,
  mean_in_cutoff = 0.8,
  max_num_gene = 3,
  n_cores = 8
)

## End(Not run)
```

featureSelectionStable

Select Spatially and Group-wise Stable Genes

Description

Highly optimized version utilizing pre-filtering and vectorized operations to bypass the computational bottleneck of gene summaries and nested loops.

Usage

```
featureSelectionStable(
  object,
  control_group_value = "0",
  check_group_diff = TRUE,
  case_group_value = "1",
  max_group_logfc_cut = 0.1,
  mean_global_cut = 0.3,
  pct_in_cut = 0.05,
  min_domain_prop = 0.8,
  max_abs_logfc_cut = 0.2,
  cv_cut = 0.3,
  max_num_gene = 50,
  use_low = TRUE,
  n_cores = 1,
  verbose = TRUE
)
```


Arguments

object	A spaCraft object.
control_group_value	Character. Value indicating the control group (default "0").
check_group_diff	Logical. Check expression difference between Control and Case (default TRUE).
case_group_value	Character. Value indicating the case group (default "1").
max_group_logfc_cut	Numeric. Maximum allowed absolute logFC between Control and Case (default 0.1).
mean_global_cut	Numeric. Minimum global mean log-expression (default 0.3).
pct_in_cut	Numeric. Minimum detection rate within a domain (default 0.05).
min_domain_prop	Numeric. Minimum proportion of domains that must meet pct_in_cut (default 0.8).
max_abs_logfc_cut	Numeric. Maximum allowed spatial logFC variance (default 0.2).
cv_cut	Numeric. Maximum allowed Robust CV across domain means (default 0.3).
max_num_gene	Integer. Maximum number of genes to return (default 50).
use_low	Logical. Use robust lower-bound logFC instead of standard logFC (default TRUE).
n_cores	Integer. Number of cores for parallel processing.
verbose	Logical. Print progress.

Value

Updated spaCraft object with `stable_genes` and `stable_gene_stats` populated.

Examples

```
## Not run:
data("mini_spaCraft_obj", package = "spaCraft")
spaCraft_obj <- featureSelectionStable(
  object = mini_spaCraft_obj,
  max_group_logfc_cut = 0.1,
  mean_global_cut = 0.1,
  min_domain_prop = 0.2,
  max_abs_logfc_cut = 0.1,
  cv_cut = 0.2,
  max_num_gene = 10,
  use_low = FALSE,
  n_cores = 8
)
## End(Not run)
```

LOR	<i>Composition Test (K-driven, Sample-Level): Naive Two-Sample or TREAT</i>
-----	---

Description

Computes per-sample $L_k = \log((y_k + \epsilon)/(b_k + \epsilon))$ and compares Case vs Control at the sample level using a Welch t-test or a Wilcoxon rank-sum test, for a single (target, reference) domain pair. Here y_k and b_k are the spot counts of the target and reference domains in sample k .

Two null modes, controlled by `null_type`:

"`treat`" (**default**) TREAT-style test (Smyth, 2009) centered on the pilot baseline $\delta_0 = \beta_1$.

- `tau = 0` (centered null): $H_0 : \hat{\delta} = \delta_0$.
- `tau > 0` (TREAT): $H_0 : |\hat{\delta} - \delta_0| \leq \tau$.

"`two_sample`" Naive two-sample test; the pilot baseline is ignored and $\delta_0 = 0$, i.e. $H_0 : \bar{\delta}_1 = \bar{\delta}_0$.

The threshold `tau` may be a numeric scalar or the string "`pilot`", in which case it is estimated from `object@pilot_data` via `estimate_lor_tau_from_pilot()` for this single (target, reference) pair.

Usage

```
LOR(
  processed_data,
  object,
  target_domain = "WM",
  reference_domain = "Layer6",
  test_method = c("t", "wilcoxon"),
  alternative = c("greater", "two.sided", "less"),
  null_type = c("treat", "two_sample"),
  tau = 0,
  pilot_tau_args = list(),
  eps = 0.5,
  verbose = TRUE
)
```

Arguments

<code>processed_data</code>	List of samples. Each element is a list with <code>coords</code> (containing <code>hat_d</code> or <code>domain</code>) and <code>meta</code> (with <code>group</code> in $\{0, 1\}$ and optionally <code>sample_id</code>).
<code>object</code>	A <code>spaCraft</code> object with <code>@params_composition</code> (and <code>@pilot_data</code> when <code>tau = "pilot"</code>).
<code>target_domain</code>	Character. Target domain label.
<code>reference_domain</code>	Character. Reference domain label.
<code>test_method</code>	"t" (Welch, default) or "wilcoxon".
<code>alternative</code>	"greater" (default), "two.sided", or "less".
<code>null_type</code>	"treat" (default) or "two_sample".

tau	Numeric scalar ≥ 0 , or the string "pilot". TREAT threshold (used only when null_type = "treat"); ignored (forced to 0) for "two_sample". If "pilot", the threshold is estimated on the fly.
pilot_tau_args	Named list of extra arguments forwarded to estimate_lor_tau_from_pilot() when tau = "pilot", e.g. list(target_prop_min = 0.12, use_noise_floor = FALSE) to anchor τ to a minimum meaningful CASE target proportion from the fitted composition baseline. object, target_domain, reference_domain, null_type, test_method, alternative, and eps are supplied automatically.
eps	Numeric > 0 . Continuity correction (default 0.5).
verbose	Logical (default TRUE).

Value

A named list with elements null_type, target, reference, K0, K1, delta_hat, delta0, delta_tilde, tau, eps, se, Z_bar0, Z_bar1, y, m, sample_id, group, stat, df, p_value, method (and stat_treat in TREAT mode). se is the Welch standard error of $\hat{\delta}$ ("t") or NA ("wilcoxon").

References

Smyth, G.K. (2009). Testing significance relative to a fold-change threshold is a TREAT. *Bioinformatics*, 25(6), 765-771.

makeCustomGeneSets	Create Custom Gene Sets for Simulation and Testing
--------------------	--

Description

Constructs customized gene sets (SVG, Null, and Spike) based on a specified target and reference domain within a selected pilot sample. Identifies the top differentially expressed genes from a pool of stable candidate genes.

Usage

```
makeCustomGeneSets(
  object,
  G_svg_base,
  G_stable,
  target_domain,
  reference_domain,
  sample_index = 1L,
  n_de = 5L,
  verbose = TRUE
)
```

Arguments

object	A spaCraft object.
G_svg_base	Character vector. Base marker genes known a priori.
G_stable	Character vector. Stable genes identified via feature selection.
target_domain	Character. The name of the target domain (e.g., "WM").

reference_domain	Character. The name of the reference domain (e.g., "Layer6").
sample_index	Integer or Character. The index or name of the pilot sample to use (default: 1L).
n_de	Integer. Number of top differentially expressed genes to extract (default: 5L).
verbose	Logical. Print progress and summary messages (default: TRUE).

Value

A list containing:

G_svg	Character vector. Union of G_svg_base and G_stable.
G_null	Character vector. Unique stable genes.
G_spike	Character vector. Top n_de differentially expressed genes.
TR_stats	Data frame. Mean expressions and differences for all tested genes.

pBANSKY

*Pilot-Guided Spatial Clustering (pBANSKY, Optimized)***Description**

Pilot-centroid-guided spatial clustering on a robust three-block BANKSY embedding (own expression C , neighbourhood mean M , and neighbourhood gradient G). Latent truth may be supplied in `coords$tilde_d` or `coords$domain`; recovered labels are written to `coords$hat_d`.

The recovered labels are statistically equivalent to a reference loop-based implementation (up to floating-point reassociation and PCA sign conventions, neither of which affects the labels), but substantially faster. The main algorithmic gains include:

1. The per-spot neighbour-mean (M) and neighbour-gradient (G) loops become sparse products with the row-normalized neighbour-weight matrix W , via $M = CW$ and $G^2 = (C \odot C)W - 2C \odot (CW) + C \odot C$ (then an elementwise square root and re-scaling).
2. PCA uses a truncated SVD (`RSpectra::svds`) for the top `n_pcs` components, falling back to `stats::prcomp` when **RSpectra** is unavailable.
3. The zero-variance row filter is fully vectorized.
4. Samples are processed in parallel across `n_cores`; the shared pilot embedding is read-only.

Usage

```
pBANSKY(
  sim_data,
  pilot_data,
  lambda,
  k_neighbors = 15,
  verbose = TRUE,
  use_G = TRUE,
  balance_mode = c("energy_proportional", "equal_split", "paper_mu"),
  mu = 1.5,
  scale_pilot_coords = TRUE,
  scale_sim_coords = TRUE,
  hard_guidance = FALSE,
```

```

    n_pcs = 20,
    do_hungarian = FALSE,
    n_cores = 1L
  )

```

Arguments

sim_data	List of simulated samples (each with counts or logcounts and a coords data frame); typically the output of <code>simulatespaCraft</code> .
pilot_data	List of pilot samples. The sample with the most domains (then the most spots) seeds the cluster centroids.
lambda	Numeric in $[0, 1]$. BANKSY spatial-mixing weight (0 = expression only, 1 = neighbourhood only).
k_neighbors	Integer. Number of spatial neighbours in the BANKSY graph. Default 15.
verbose	Logical. Print progress. Default TRUE.
use_G	Logical. Include the neighbourhood-gradient (G) block. Default TRUE.
balance_mode	Character. How the spatial budget is split between the M and G blocks: one of "energy_proportional" (default), "equal_split", or "paper_mu".
mu	Numeric. Gradient down-weighting, used only when <code>balance_mode = "paper_mu"</code> . Default 1.5.
scale_pilot_coords	Logical. Min-max scale the pilot coordinates to the unit square before graph construction. Default TRUE.
scale_sim_coords	Logical. Min-max scale the simulated coordinates. Default TRUE.
hard_guidance	Logical. If TRUE, assign each spot to its nearest pilot centroid instead of running k-means. Default FALSE.
n_pcs	Integer. Number of principal components of the joint embedding. Default 20.
do_hungarian	Logical. If TRUE and latent truth is available, align recovered labels to truth via the Hungarian algorithm. Default FALSE.
n_cores	Integer. Samples processed in parallel (clamped to 1 on Windows). Default 1.

Value

A list of processed samples. Each element carries the input counts, a coords data frame with the recovered domain in `hat_d` (and a per-spot `is_correct` flag when truth is available), plus clustering diagnostics (`global_accuracy`, `n_domains`).

plot, spaCraft, missing-method

Plot Spatial Domains for a spaCraft Object

Description

A unified plotting method for spaCraft objects. Visualizes the spatial distribution of domains for both pilot data and synthetic data. It automatically extracts unified domain levels so that colours are consistent across all plots.

Usage

```
## S4 method for signature 'spaCraft,missing'
plot(
  x,
  y,
  type = c("pilot", "synthetic_raw", "synthetic_mapped"),
  color_by = c("hat_d", "domain"),
  n_per_group = 3,
  ncol = 3,
  sample_ids = NULL,
  cex = 1,
  alpha = 0.9,
  title = NULL,
  subtitle = NULL,
  colors = NULL,
  reverse_y = FALSE,
  ...
)
```

Arguments

<code>x</code>	A <code>spaCraft</code> object.
<code>y</code>	Not used. Required for S4 generic compatibility.
<code>type</code>	Character. Which data to plot: "pilot" (ground truth), "synthetic_raw" (before rearrangement, continuous), or "synthetic_mapped" (after rearrangement, KD-tree snapped).
<code>color_by</code>	Character. For synthetic data, colour by the recovered domains ("hat_d", default) or by truth ("domain").
<code>n_per_group</code>	Integer. Number of samples to show per group. Default 3.
<code>ncol</code>	Integer. Number of columns in the facet wrap. Default 3.
<code>sample_ids</code>	Optional character vector of specific sample IDs to plot.
<code>cex</code>	Numeric. Point-size expansion factor. Default 1.0.
<code>alpha</code>	Numeric. Point transparency in $[0, 1]$. Default 0.9.
<code>title</code>	Character. Custom plot title; if NULL, a default is used.
<code>subtitle</code>	Character. Custom subtitle; if NULL, a default is used.
<code>colors</code>	Character vector of colours. If NULL, uses the "Set1" palette.
<code>reverse_y</code>	Logical. If TRUE, reverses the y-axis to match image coordinates. Default FALSE.
<code>...</code>	Additional arguments (currently ignored).

Value

A `ggplot` object.

plot.spaCraftTuning	<i>Plot Method for spaCraftTuning</i>
---------------------	---------------------------------------

Description

Plot Method for spaCraftTuning

Usage

```
## S3 method for class 'spaCraftTuning'
plot(x, ...)
```

Arguments

x	An object of class spaCraftTuning.
...	Additional arguments to be passed to plotting functions.

plotPowerCurveLOR	<i>Plot Composition (LOR) Power / Size Curve</i>
-------------------	--

Description

Smooths the empirical rejection rate (= power for effect > 0, = size at effect = 0) against K, one curve per composition effect size, with a dashed reference line at the nominal level alpha. Uses a shape-constrained additive model (monotone-increasing in K, scam, bs = "mpi") when **scam** is installed and there are >= 4 distinct K; otherwise falls back to monotone isotonic interpolation.

Usage

```
plotPowerCurveLOR(summary_df, alpha = 0.05, k_spline = 5)
```

Arguments

summary_df	Data frame from evaluatePowerLOR.
alpha	Numeric. Nominal level (reference line + title).
k_spline	Integer. Spline basis dimension (capped at #unique K).

Value

A ggplot object.

See Also

[evaluatePowerLOR](#)

plotPowerCurveSaLFC	<i>Plot SCAM-Smoothed SaLFC Power Curve (soft power)</i>
---------------------	--

Description

Smooths the empirical SOFT-power surface ($\text{power} = \overline{n_{TP}/n_{DE}}$) with a shape-constrained additive model (monotone-increasing in K per effect size, `bs = "mpi"`). Falls back to monotone isotonic interpolation with a warning if **scam** is unavailable or there are < 4 distinct K. Each anchor is labelled with the fitted power and the mean false-discovery proportion (FDR), which are reported on independent axes. A dashed reference line marks `target_power`.

Usage

```
plotPowerCurveSaLFC(summary_df, alpha = 0.05, k_spline = 5, target_power = 0.8)
```

Arguments

<code>summary_df</code>	Data frame from <code>evaluatePowerSaLFC</code> .
<code>alpha</code>	Numeric. Significance level (title only).
<code>k_spline</code>	Integer. Spline basis dimension (capped at #unique K).
<code>target_power</code>	Numeric in (0,1). Dashed reference line.

Value

A ggplot object.

See Also

[evaluatePowerSaLFC](#)

print.powerLOR	<i>Print a LOR composition power grid (compact)</i>
----------------	---

Description

Print a LOR composition power grid (compact)

Usage

```
## S3 method for class 'powerLOR'
print(x, digits = 3, ...)
```

Arguments

<code>x</code>	A powerLOR object.
<code>digits</code>	Rounding for the rate matrix.
<code>...</code>	Ignored.

Value

`x`, invisibly.

print.powerSaLFC	<i>Print a SaLFC power grid (compact, soft power)</i>
------------------	---

Description

Print a SaLFC power grid (compact, soft power)

Usage

```
## S3 method for class 'powerSaLFC'
print(x, digits = 3, ...)
```

Arguments

x	A powerSaLFC object.
digits	Rounding for the power matrix.
...	Ignored.

Value

x, invisibly.

rearrangeSyntheticToPilot	<i>Rearrange Synthetic Spots onto Pilot Domain Geometry (Per-Domain OT)</i>
---------------------------	---

Description

Reshapes blob-like simulated domains so that each domain occupies the same spatial region as the corresponding domain in a reference pilot sample, via a two-stage per-domain optimal transport (exact 2D-Gaussian affine alignment, then bijective nearest-neighbour snapping onto real pilot spots).

Composition-aware mode (`boundary_shift = TRUE`). The default per-domain map sends each synthetic domain onto the SAME domain's pilot footprint, so an increased case-group target proportion is rendered as higher *density* inside the fixed (control) pilot target footprint rather than a larger *area*. When `boundary_shift = TRUE` (with `target_domain` and `reference_domain` supplied), the target and reference pilot spots are pooled and re-partitioned per section so that the target **territory area fraction matches that section's own target proportion** $f = n_{target} / (n_{target} + n_{reference})$: the target region is grown contiguously into the nearest reference spots until it holds an f fraction of the pool. Synthetic target spots are then placed in this (enlarged, for case) territory and reference spots in the remainder. Labels and per-domain counts are untouched (so the LOR statistic is unchanged); only the spatial footprint reflects the effect. Control sections (baseline f) look like the pilot; case sections show the target band expanding into the reference. All other domains map by the standard per-domain OT.

Usage

```
rearrangeSyntheticToPilot(
  synthetic_data,
  pilot_data,
  match_by = c("hat_d", "domain"),
  jitter_frac = 0.4,
  k_nn = 80L,
  seed = 1L,
  verbose = TRUE,
  boundary_shift = FALSE,
  target_domain = NULL,
  reference_domain = NULL
)
```

Arguments

<code>synthetic_data</code>	List of simulated samples; each has coords with x, y, the <code>match_by</code> label, and <code>meta\$group/meta\$sample_id</code> .
<code>pilot_data</code>	List of pilot samples; each has coords with x, y, domain, group, <code>sample_id</code> .
<code>match_by</code>	"hat_d" (default) or "domain".
<code>jitter_frac</code>	Numeric. Oversampling jitter as a fraction of local spacing. Default 0.4.
<code>k_nn</code>	Integer. Candidate neighbours for the greedy assignment. Default 80.
<code>seed</code>	Integer base seed (global RNG restored on exit). Default 1.
<code>verbose</code>	Logical. Default TRUE.
<code>boundary_shift</code>	Logical. If TRUE, render the target/reference composition difference as a moving boundary (see Description). Default FALSE (original per-domain behaviour, unchanged).
<code>target_domain, reference_domain</code>	Character. The domain pair whose boundary moves when <code>boundary_shift = TRUE</code> (e.g. "WM" and "Layer6"). Must match the labels in the <code>match_by</code> column.

Value

The input list with `coords$x/coords$y` replaced by the rearranged coordinates; originals kept as `x_raw/y_raw`, and `matched_pilot_id` records the reference section.

See Also

[pBANSKY](#), [simulatespaCraft](#)

Description

Tests for a differential Target-vs-Reference log-fold-change between two groups of spatial samples, using ground-truth domain labels. Per sample, the statistic $z = \bar{x}_T - \bar{x}_R$ is formed together with its spatial sampling variance τ^2 (derived from an exponential covariance with the per-gene parameters in `object@params_expression`). Across samples, the biological variance is estimated, optionally moderated via Smyth-style empirical Bayes (`limma::squeezeVar`), and a TREAT-style threshold test against `lfc_threshold` is performed with Satterthwaite degrees of freedom. A rank-based (Wilcoxon) alternative is also available.

Usage

```
SaLFC(
  processed_data,
  object,
  genes,
  target_domain = "WM",
  reference_domain = NULL,
  domain_col = c("hat_d", "domain"),
  rho_TR = 0,
  adjust_method = c("bonferroni", "BH"),
  test_method = c("wilcoxon", "t"),
  alternative = c("greater", "less", "two.sided"),
  lfc_threshold = 0,
  pilot_threshold_args = list(),
  max_spots_per_domain = 5000L,
  seed = 2026L
)
```

Arguments

<code>processed_data</code>	List of samples (e.g. recovered synthetic data). Each element must contain counts (genes x spots) and coords with columns x, y, and the label column selected by <code>domain_col</code> (<code>hat_d</code> and/or <code>domain</code>); optionally <code>meta\$group</code> and <code>meta\$sample_id</code> .
<code>object</code>	A <code>spaCraft</code> object, used for per-gene spatial parameters (<code>theta</code>) via <code>object@params_expression</code> .
<code>genes</code>	Character vector of genes to test.
<code>target_domain</code>	Character. Target domain label.
<code>reference_domain</code>	Character or NULL. Reference domain; if NULL, all non-target spots are used as the reference.
<code>domain_col</code>	Character. Which coords column defines the target and reference domains: <code>"hat_d"</code> (recovered clusters, default) or <code>"domain"</code> (ground truth). Falls back to <code>"domain"</code> when the requested column is absent (e.g. pilot data, which carries only ground-truth labels).
<code>rho_TR</code>	Numeric. Assumed Target-Reference biological correlation (< 1).
<code>adjust_method</code>	Multiple-testing correction: <code>"BH"</code> (FDR, default) or <code>"bonferroni"</code> (FWER).
<code>test_method</code>	Group comparison on per-sample z_{hat} : <code>"wilcoxon"</code> or <code>"t"</code> .
<code>alternative</code>	Test direction: <code>"greater"</code> (default), <code>"less"</code> , or <code>"two.sided"</code> .

lfc_threshold Numeric scalar, named numeric vector keyed by gene, or the string "pilot". If a named vector, an optional "global_tau" attribute supplies the fallback for missing/non-finite entries. If "pilot", the threshold is estimated on the fly from `object@pilot_data` via `estimate_lfc_threshold_from_pilot()` and then used for the test.

pilot_threshold_args Named list of extra arguments forwarded to `estimate_lfc_threshold_from_pilot()` when `lfc_threshold = "pilot"` (e.g. `list(q = 0.75, min_tau = 0.05)`). `object`, `genes`, `target_domain`, `reference_domain`, `rho_TR`, `adjust_method`, `test_method`, and `alternative` are supplied automatically but may be overridden here.

max_spots_per_domain Integer. Per-domain spot cap; domains larger than this are randomly down-sampled (bounds the $O(n^2)$ distance matrices). The global RNG state is restored on exit.

seed Integer base seed for reproducible down-sampling.

Value

A data.frame of per-gene moderated statistics and adjusted p-values, ordered by `padj` then `p_value`. An empty data.frame is returned when no gene has sufficient data.

See Also

[evaluatePowerSaLFC](#), [estimate_lfc_threshold_from_pilot](#)

Examples

```
## Not run:
data("mini_obj_fitted", package = "spaCraft")
data("mini_custom_genes", package = "spaCraft")

## One-point SaLFC power calculation (single design point: K = 5, DE_lfc = 0.3):
## simulate -> recover domains -> rearrange -> SaLFC -> rejection rate.
sim <- simulateSpaCraft(mini_obj_fitted, n_sample_per_group = 5,
  scenario_settings = list(DE_lfc = 0.3), target_domain = "WM",
  genes_to_simulate = mini_custom_genes$G_svg,
  de_genes = mini_custom_genes$G_spike, seed_base = 1)

syn <- pBANSKY(sim, mini_obj_fitted@pilot_data, lambda = 0.5, do_hungarian = TRUE)
syn <- setNames(lapply(syn, function(s) list(counts = s$counts, coords = s$coords,
  meta = data.frame(sample_id = s$sample_id, group = s$group))),
  vapply(syn, function(s) s$sample_id, character(1)))
syn <- rearrangeSyntheticToPilot(syn, mini_obj_fitted@pilot_data, match_by = "hat_d")

res <- SaLFC(syn, mini_obj_fitted,
  genes = unique(c(mini_custom_genes$G_null, mini_custom_genes$G_spike)),
  target_domain = "WM", reference_domain = "Layer6", domain_col = "hat_d",
  lfc_threshold = "pilot")

## empirical power at this point = fraction of true spike-ins at FDR <= 0.05
mean(res$padj[res$gene %in% mini_custom_genes$G_spike] <= 0.05)

## End(Not run)
```

simulatespaCraft	<i>Simulate Spatial Transcriptomics Data (Hybrid Mode, Optimized)</i>
------------------	---

Description

Optimized rewrite of `simulatespaCraft`. Statistically identical to the previous version under the same `seed_base`, but substantially faster. The main algorithmic gains include:

- The conditional-texture (`eta_cond`) KNN structure is built once per domain per sample and reused across all genes.
- Sample/group invariants are hoisted out of the per-sample and per-gene loops.
- The per-spot domain-mean lookup is fully vectorized.

Usage

```
simulatespaCraft(
  object,
  n_sample_per_group = 10,
  scenario_settings = list(DE_lfc = 0, target_prop_case = NULL, delta_pp = NULL, kappa_m
    = 1, kappa_bio = 1, delta_rho = 1, lambda_cond = 0.1),
  target_domain = NULL,
  genes_to_simulate = NULL,
  de_genes = NULL,
  verbose = TRUE,
  n_cores = 4,
  gp_rank = 20L,
  gp_jitter = 1e-08,
  spatial_mode = c("original", "basis"),
  graph_k = 20L,
  cond_k_nn = 20L,
  basis_rank = 25L,
  basis_kernel = c("gaussian", "exponential"),
  basis_knot_method = c("sample", "grid"),
  basis_seed = 1L,
  lambda_cloud = NULL,
  seed_base = NULL
)
```

Arguments

<code>object</code>	A <code>spaCraft</code> object.
<code>n_sample_per_group</code>	Integer. Number of synthetic samples to generate per group.
<code>scenario_settings</code>	List. Scenario configurations including <code>DE_lfc</code> , <code>target_prop_case</code> , <code>kappa_m</code> , <code>kappa_bio</code> , <code>delta_rho</code> , and <code>lambda_cond</code> .
<code>target_domain</code>	Character. Target domain name for differential effects.
<code>genes_to_simulate</code>	Character vector. Genes to simulate.

de_genes	Character vector. Subset of genes that will have DE effects applied.
verbose	Logical. Print progress.
n_cores	Integer. Number of cores for parallel execution.
gp_rank	Integer. Rank for Gaussian Process (legacy).
gp_jitter	Numeric. Jitter to ensure positive definiteness.
spatial_mode	Character. "original" or "basis".
graph_k	Integer. KNN for the whole-tissue spatial graph.
cond_k_nn	Integer. KNN for the pilot-guided conditional texture.
basis_rank	Integer. Basis rank for spatial mode "basis".
basis_kernel	Character. "gaussian" or "exponential".
basis_knot_method	Character. "sample" or "grid".
basis_seed	Integer. Seed for basis knot sampling.
lambda_cloud	Optional numeric in $[0, 1]$. Used for FGKMM simulation.
seed_base	Optional integer for reproducible simulations.

Value

A list of simulated synthetic samples.

spaCraft-class	<i>spaCraft Class</i>
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Description

S4 class for multi-sample spatial transcriptomics power analysis. Stores the pilot data, the hierarchical model parameters fitted from it, the simulated synthetic data, and the downstream testing results.

Slots

pilot_data list. Pilot samples; each element contains counts (genes x spots), coords (x, y, domain), group (0/1), and sample_id.

params_geometry list. Hierarchical FGKMM parameters (centroids, shapes, pooling info).

params_expression list. Hierarchical NNGP parameters (mean, covariance, sigma_bio).

params_composition list. Beta-Binomial parameters for domain proportions.

synthetic_data list. Simulated/recovered samples produced by the simulation and clustering pipeline. Each element holds counts, coords, and meta.

testing_result list. Endpoint testing results, keyed by endpoint (e.g., salfc, composition).

summary.powerLOR	<i>Summarize a LOR composition power analysis (lm-style report)</i>
------------------	---

Description

Produces an interpretive summary: a rejection-rate table over K x effect, the minimum K reaching target_power per (non-null) effect (grid + continuous estimate), and the empirical size at the null.

Usage

```
## S3 method for class 'powerLOR'
summary(object, target_power = 0.8, ...)
```

Arguments

object	A powerLOR object.
target_power	Numeric in (0,1).
...	Ignored.

Value

An object of class summary.powerLOR.

summary.powerSaLFC	<i>Summarize a SaLFC power analysis (lm-style report, soft power)</i>
--------------------	---

Description

Produces an interpretive summary on the SOFT-power metric (power = mean(power_raw) = mean(n_TP/n_DE)): a power table over K x DE_lfc, the minimum K reaching target_power per effect (grid value plus a monotone-interpolated continuous estimate), and a separate FDR-control diagnostic. The all-or-nothing power_strict is retained in \$by_cell for reference only.

Usage

```
## S3 method for class 'powerSaLFC'
summary(object, target_power = 0.8, ...)
```

Arguments

object	A powerSaLFC object.
target_power	Numeric in (0,1).
...	Ignored.

Value

An object of class summary.powerSaLFC (a list with \$by_cell, \$power, \$se, \$min_k, \$fdr, \$design); printed as a report. \$power is the soft-power matrix.

summary.spaCraftTuning

Summary Method for spaCraftTuning

Description

Summary Method for spaCraftTuning

Usage

```
## S3 method for class 'spaCraftTuning'
summary(object, ...)
```

Arguments

object	An object of class spaCraftTuning.
...	Additional arguments affecting the summary produced.

syntheticData

Get or Set Synthetic Data

Description

Accessor and replacement methods for the synthetic_data slot of a spaCraft object.

Usage

```
syntheticData(object)

## S4 method for signature 'spaCraft'
syntheticData(object)

syntheticData(object) <- value

## S4 replacement method for signature 'spaCraft'
syntheticData(object) <- value
```

Arguments

object	A spaCraft object.
value	A named list of simulated samples to store.

Value

syntheticData returns the stored list; the replacement method returns the updated spaCraft object.

testingResult	<i>Get or Set Testing Results</i>
---------------	-----------------------------------

Description

Accessor and replacement methods for the testing_result slot of a spaCraft object.

Usage

```
testingResult(object)

## S4 method for signature 'spaCraft'
testingResult(object)

testingResult(object) <- value

## S4 replacement method for signature 'spaCraft'
testingResult(object) <- value
```

Arguments

object	A spaCraft object.
value	A named list of endpoint results to store.

Value

testingResult returns the stored list; the replacement method returns the updated spaCraft object.

tuneSpaDesignLambdas	<i>Stability- and Validity-Driven Hyperparameter Tuning</i>
----------------------	---

Description

Tunes the spatial hyperparameters of a spaCraft object so that the resulting simulation behaves as a *faithful and stable power oracle*. The recovery weight lambda_p (pBANKSY) is tuned once by maximizing clustering reproducibility (ARI), exactly as before. The conditional-texture weight lambda_cond is tuned with a regularized-noncentrality objective that fixes two defects of a pure $Var(T)$ -minimization:

1. **Objective.** Power is governed by the noncentrality $\delta = E[T]/sd[T]$, not by $sd[T]$ alone. Minimizing variance can be achieved by attenuating the signal, which collapses power. The objective is therefore a regularized noncentrality (equivalently the reciprocal of a floored coefficient of variation),

$$\hat{\delta}(\lambda) = \bar{T}/(sd(T) + \varepsilon s_0),$$

which is scale-free and cannot be gamed by uniform shrinkage. Both endpoints are placed on the *same* standardized (t-statistic) footing: the SaLFC endpoint uses `mean(abs(t_stat))` and the LOR endpoint uses the standardized statistic `LOR()$stat` (not the raw `delta_hat`).

2. **Effect ordering.** The default `lambda_cond_selection = "per_effect"` selects the pointwise-optimal weight at every effect size, $\lambda^*(e) = \arg \max_{\lambda} \widehat{\delta}(e, \lambda) w(\lambda)$, where $w(\lambda)$ is the null-calibration weight below. Because the generator's noncentrality is monotone in effect size for each fixed λ , the pointwise maximum $\max_{\lambda} \widehat{\delta}(e, \lambda)$ is itself monotone in e in expectation, so per-effect selection does not sacrifice the effect ordering; residual non-monotonicity is Monte-Carlo noise and is removed by isotonizing (PAVA) the *reported* noncentrality (`SNR_iso`). The per-effect `lambda_cond` is stored in `by_effect` and consumed downstream (e.g. [evaluatePowerSaLFC](#), [evaluatePowerLOR](#)), which already match `lambda_cond` to each effect size. Setting `lambda_cond_selection = "shared"` recovers the previous behaviour: a single `lambda_cond` is selected jointly across the effect grid by maximizing the worst-case (minimum-over-effects) calibrated noncentrality, trading a little per-effect optimality for lower selection variance.

A null-calibration guard (`calibration_guard`) further down-weights any `lambda_cond` whose empirical type-I error under the matched null (SaLFC: `DE_lfc = 0`, no DE genes; LOR: `target_prop_case = p0`, the pilot baseline proportion) is inflated, so noncentrality is never bought by biasing the generator.

Usage

```
tuneSpaDesignLambdas(
  object,
  target_domain,
  reference_domain,
  G_DE,
  scenario_H1,
  lambda_p_grid = seq(0.2, 0.8, 0.1),
  lambda_c_grid = seq(0.2, 0.8, 0.1),
  lambda_cond_selection = c("per_effect", "shared"),
  B = 10,
  K_syn = 10,
  n_cores = 1,
  seed = 2026,
  verbose = TRUE,
  eps_floor = 0.001,
  calibration_guard = TRUE,
  alpha = 0.05,
  calib_mult = 2
)
```

Arguments

<code>object</code>	A <code>spaCraft</code> object.
<code>target_domain</code>	Character. Target domain name.
<code>reference_domain</code>	Character. Reference domain name.
<code>G_DE</code>	Character vector. Genes with a true DE effect (H1 scenario).
<code>scenario_H1</code>	List. Joint scenario settings for H1. <code>DE_lfc</code> and <code>target_prop_case</code> may each be a scalar or a numeric vector; when a vector, <code>lambda_cond</code> is evaluated at every value and (depending on <code>lambda_cond_selection</code>) either a per-effect or a single shared value is selected across the grid.
<code>lambda_p_grid</code>	Numeric vector. Grid of <code>lambda_p</code> to test.

lambda_c_grid	Numeric vector. Grid of lambda_cond to test.
lambda_cond_selection	Character. How the conditional-texture weight is chosen across the effect grid. "per_effect" (default) returns the pointwise-optimal (calibrated) lambda_cond at each effect size, stored in by_effect\$lambda_cond and matched per effect by the downstream power functions; "shared" returns a single value maximizing the worst-case calibrated noncentrality across effects (the previous behaviour).
B	Integer. Number of replicates per grid point (Default: 10).
K_syn	Integer. Number of samples per group during tuning (Default: 10).
n_cores	Integer. Number of cores for simulation.
seed	Integer. Random seed for reproducible tuning. Default is 2026.
verbose	Logical. Prints progress.
eps_floor	Numeric. Denominator floor multiplier ε for the regularized noncentrality (Default: 1e-3). Stabilizes the $E[T] \rightarrow 0$ regime (relevant for LOR near the null).
calibration_guard	Logical. If TRUE (default) compute the null-calibration weight and down-weight inflated lambda_cond.
alpha	Numeric. Nominal significance level for the calibration guard.
calib_mult	Numeric > 1. Tolerance multiplier: the calibration weight is ≈ 1 while the empirical type-I error stays below $\alpha * \text{calib_mult}$ and decays smoothly beyond it.

Value

An object of class spaCraftTuning. optimal_params has one entry per endpoint (SaLFC, LOR); each contains the shared lambda_p, a representative scalar lambda_cond (the value at the median effect), a by_effect data frame (the selected lambda_cond *per effect size*, realized noncentrality SNR, its isotonic projection SNR_iso, and the per-effect calibration weight calib_weight), and shared_lambda_cond (the single shared weight when lambda_cond_selection = "shared", otherwise NA). summary_stage2 holds the per-endpoint noncentrality surfaces (with mean_T, sd_T, SNR, and Var_T for reference). diagnostics stores the calibration table.

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